Examen LFA

1. Fie $N = (\{0,1\}, \{a,b\}, \{(0,a,0), (0,b,0), (0,b,1), (1,a,1), (1,a,0)\}, 0, \{1\})$. Scrieti un automat determinist A astfel incat $L(N) = L(A)$.

   Solutie: just do it.

2. Scrieti o gramatica G in Forma Normala Cholmsky, pentru limbajul $\{a^n b^n \mid n \geq 1\}$.

   Solutie: $S \to ZA$, $A \to SU \mid 1$, $Z \to a$, $U \to b$

3. Demonstrati folosind Lema de Pompare ca limbajul $L = \{w_1 w_2 \in \{0,1\}^+ \mid \#_1(w_1) = 2 \cdot \#_0(w_1)$ si $\#_1(w_2) = 2 \cdot \#_0(w_2)\}$ nu este regulat.

   Solutie: $w_n = 0^n 1^{2n} 011$ *si* $i = 0$. *Atunci* $xy^i z = 0^{n+(i-1)b} 1^{2n} 011$. *Pentru* $i = 0, w_n = 0^{n-b} 1^{2n} 011$, *cu* $b \neq 0$. *Cum* $\#_0(w_n)$ *este cel mult* n *iar* $\#_1(w_n) = 2n+2$, *nu putem imparti* w_n *in doua cuvinte* w_1 *si* w_2 *astfel incat* $w_1 w_2 \in L$.

4. Scrieti o Masina Turing care accepta limbajul $\{a^n b^n \mid n \geq 1\}$.

   Solutie:

	a	b	A	B	$\#$
0	$1, A/R$			$0, B/R$	$3, \#/H$
1	$1, a/R$	$2, B/L$		$1, B/R$	
2	$2, a/L$		$0, A/R$	$2, B/L$	
3					

5. Aratati ca implicatia: $''\forall L : L_1 \cdot L_2$ regulat $\implies$ L_1 regulat si L_2 regulat $''$ este falsa.

   Solutie: $L_1 = \{0^n 1^m \mid n > m \geq 0\}$ *si* $L_2 = L(1^*)$. $L_1 L_2 = L(0^* 1^*)$.

6. Scrieti o gramatica neambigua care genereaza acelasi limbaj ca gramatica: $S \to AS \mid \epsilon$, $A \to A1 \mid 0A1 \mid 01$.

   Solutie: $S \to S1 \mid 0S1 \mid 01 \mid \epsilon$

7. Definiti un APD care accepta limbajul $\{w_1 w_2 \in \{a,b\}^* \mid \#_a(w_1) = \#_b(w_2)\}$